

TED: Abusing Tunnel Hosts and IPv6 Extension Headers for Pulsing DoS Attacks

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Abstract

IP tunneling mechanisms are widely deployed to facilitate the Internet’s transition from IPv4 to IPv6, yet their security implications remain insufficiently scrutinized. In this paper, we present TED, a novel pulsing denial-of-service attack that exploits structural vulnerabilities in IP tunneling and IPv6 Extension Headers (EHs) processing. Unlike prior reflection attacks that depend on application-layer services, TED operates entirely at the network layer, requiring no victim interaction, prolonged traffic accumulation, or protocol-specific dependencies. Delay lines are constructed by TED through nested EHs, capitalizing on the tunnel hosts’ blind forwarding logic and lack of deep packet inspection. These artificial delay lines allow attackers to solve the send-time schedule problem, converging asynchronous, low-rate traffic into a destructive, high-magnitude pulse at the victim. Our Internet-wide measurement identified over 1.9 million vulnerable tunnel hosts acting as unwitting relays, including hosts within critical satellite infrastructure. Evaluation results demonstrate that TED achieves normalized pulse amplitude exceeding $180\times$ while successfully evading existing low-rate and pulsing attack detection mechanisms based on specific application layer protocols. We discuss the root causes of these transitional vulnerabilities and possible mitigation strategies for network operators and equipment vendors.

1 Introduction

Denial-of-service (DoS) attacks remain a persistent and highly disruptive threat to Internet services. By exhausting critical resources, including bandwidth, CPU, and memory, attackers can deny legitimate users access to online services. Classic DoS attacks typically rely on sustained flooding, often using reflection and amplification via services such as DNS and NTP [14, 54, 64]. Because these floods exhibit continuous high-traffic signatures, they are increasingly detectable by modern defense systems [1, 6, 11, 21, 27, 39, 61–63].

Consequently, attackers have shifted toward distributed and more evasive strategies. Distributed denial-of-service (DDoS)

attacks, which aggregate traffic from many sources, such as botnets comprised of compromised devices [38], can overwhelm targets and cause substantial service disruption and financial loss. Moreover, to further reduce detectability, attackers increasingly adopt pulsing denial-of-service (PDoS) strategies [31, 45, 47, 48, 57]. Rather than continuous flooding, PDoS attacks generate short, high-intensity bursts that concentrate traffic into narrow time windows while maintaining a low average rate, thereby evading threshold-based detection.

Prior work has demonstrated pulsing mechanisms such as temporal lensing [57], which exploits latency heterogeneity across network paths, and DNSBomb [45], which leverages resolver timeouts and aggregation effects. Concurrently, these lines of work typically rely on either specific application-layer middleboxes (e.g., open DNS resolvers or CDNs) or fixed path properties to induce traffic convergence and pulse concentration. However, these prior approaches largely overlook the potential of network-layer tunnel nodes as a programmable alternative for traffic convergence.

At the same time, the Internet is undergoing a transition from IPv4 to IPv6. To bridge the two protocols, tunneling mechanisms (e.g., 6to4 [10] and GRE [44]) are widely deployed [5]. Recent studies have shown that many tunnel hosts behave as open relays and can be abused as forwarding proxies [5]. Moreover, IPv6 introduces Extension Headers to support fragmentation, routing, and security options. However, inconsistent EH processing across vendors and networks creates a complex and underexplored attack surface for malicious traffic manipulation [9, 26, 42].

In this paper, we propose TED, a novel pulsing attack that operates entirely at the network layer and requires neither prolonged traffic accumulation nor protocol-specific dependencies. TED exploits the key insight that tunnel hosts often do not perform deep packet inspection (DPI) of nested tunnel packets. By encapsulating IPv6 tunnel packets with crafted Extension Headers, the attacker can create programmable *delay lines* in the network, precisely controlling per-packet forwarding latency and routing path of each packet, causing traffic to converge at the victim within a chosen time win-

dow. Unlike prior approaches that rely on passive aggregation effects [57], TED actively schedules packets. By combining multi-hop forwarding paths with delays spanning tens of microseconds to seconds, the attacker solves a send-time schedule problem in which packets sent asynchronously over an extended period arrive nearly simultaneously at the victim, producing a high-magnitude pulse.

Our measurement study of the IPv4/IPv6 address space reveals a vast attack surface: we identified over 1.9 million vulnerable tunnel hosts across six protocol types. Leveraging these hosts, TED achieves an normalized pulse amplitude (PA) exceeding $180\times$ with only seconds of preparation time. This significantly outperforms prior methods [57] that rely on intrinsic path latency differences to convert low-bandwidth continuous traffic into short-burst, high-bandwidth traffic. Crucially, because TED requires no interaction with application-layer services, it remains impervious to upper-layer defense mechanisms.

In summary, we make the following contributions:

- **New Attack Vector:** We propose a novel attack mechanism that combines IP tunneling protocols with Extension Headers to manufacture programmable path delays, effectively transforming benign tunnel hosts into components of a massive DDoS weapon.
- **System Design:** We design a comprehensive attack framework comprising topology discovery, latency measurement, source address spoofing, and a precise packet scheduler. Furthermore, we introduce a path pruning algorithm to prevent intermediate node congestion, thereby ensuring stealth during the attack accumulation phase.
- **High-Impact Evaluation:** Through controlled experiments in real-world environments, we demonstrate that TED can concentrate low-rate traffic into high-volume pulses with a normalized pulse amplitude of up to $180\times$. In contrast, existing normal PDoS attacks typically achieve only around $10\text{--}15\times$ temporal concentration.
- **Availability:** To foster future research and the development of defenses, we have released our measurement dataset and Proof-of-Concept attack code¹.

The remainder of this paper is organized as follows: Section 2 provides background on tunneling, IPv6 EHs, and PDoS attacks. Section 3 presents the EH processing analysis. Section 4 details the design and implementation of TED. Section 5 evaluates the attack surface and real-world impact. Section 6 discusses root causes and mitigation strategies. Section 7 surveys related work, and Section 8 concludes.

2 Background

This section first introduces IP tunnel protocols and IPv6 extension headers, and then explains the basic principles underlying PDoS attacks.

2.1 IP Tunneling Mechanisms

IP tunneling is a core networking mechanism in which an IP packet is encapsulated within a new outer IP header [5, 30]. Encapsulation allows traffic to traverse intermediate networks with different protocol support or addressing constraints, effectively forming a virtual point-to-point link while shielding the inner packet from intermediate routers. At the tunnel host, the outer header is removed, and the original packet is forwarded toward its destination. Tunneling underpins a range of deployments, including IPv4/IPv6 transition, interconnection of disjoint networks (e.g., mobile or satellite), and VPN connectivity over public infrastructure [43]. This study considers tunneling protocols that, by default, provide neither authentication nor encryption. Under such configurations, a tunnel host that accepts encapsulated traffic from arbitrary sources acts as an open relay; we term such hosts *vulnerable hosts*. In networks without source address validation, attackers can abuse these hosts to spoof source addresses or relay DoS traffic. In this section, we examine three representative classes of tunneling protocols.

IPIP/IP6IP6. Defined in RFC 2003 [55] and RFC 2473 [15] respectively, these protocols represent the most fundamental approach to encapsulation. IP in IP (IPIP) embeds an IPv4 packet directly within another IPv4 header, while IPv6 in IPv6 (IP6IP6) performs the same operation for IPv6. Designed for minimalism, they lack intermediate headers or cryptographic features, making them ideal for low-overhead connectivity between disjoint networks of the same IP version. But it also increases the risk of abuse when deployed without strict access restrictions.

GRE/GRE6. Generic Routing Encapsulation (GRE) is standardized in RFC 2784 [44], and its IPv6 variant (GRE6) is specified in RFC 7676 [56]. Compared to IPIP, GRE introduces an additional GRE header between the outer transport header and the payload. The header includes a protocol type field that supports protocol-independent encapsulation, and optional fields that enable basic tunnel identification and traffic separation. However, GRE provides neither encryption nor strong authentication, and its security therefore relies on external control mechanisms. GRE6 extends the same encapsulation model to IPv6 transport networks, enabling other payloads to traverse IPv6-only backbones.

6in4/4in6. The IPv6 in IPv4 (6in4) and IPv4 in IPv6 (4in6) protocols are IPv4/IPv6 transition mechanisms that enable connectivity across heterogeneous networks. The 6in4 protocol, specified in RFC 4213 [24], encapsulates an IPv6 packet within an outer IPv4 header by setting the IPv4 Protocol field

¹<https://github.com/THUname/TED>

to 41, allowing IPv6 traffic to traverse IPv4-only networks during early transition. In contrast, 4in6 (RFC 2473 [15]) encapsulates a complete IPv4 packet within IPv6 by setting the IPv6 *Next Header* field to 4; at the tunnel hosts, the outer IPv6 header is removed, and the inner IPv4 packet is forwarded according to standard IPv4 routing, preserving IPv4 reachability over IPv6-only core networks. An important automatic variant of 6in4 is 6to4 (RFC 3056 [10]), which derives an IPv6 prefix (2002::/16) from a host’s public IPv4 address to reduce configuration overhead. Interworking with the native IPv6 network depends on publicly reachable relay routers. Although RFC 3964 recommends strict source address validation (SAV) to mitigate spoofing [53], such checks are often missing in practice, leaving 6to4 deployments vulnerable to abuse [29].

Security Implications. While essential for connectivity, many tunneling hosts (e.g., border routers) operate as *Open Relays*. As noted in recent studies [5, 29], these hosts can forward encapsulated traffic from arbitrary sources to arbitrary destinations without strict authentication or SAV. This indiscriminate forwarding behavior makes them ideal reflectors for attackers to exploit.

2.2 IPv6 Extension Headers

Unlike IPv4, which uses a variable-length header to accommodate optional fields, IPv6 adopts a fixed 40-byte base header. To preserve extensibility without increasing the complexity of the base header, IPv6 introduces extension headers. These optional headers are placed between the IPv6 base header and the upper-layer transport protocol (e.g., TCP or UDP). Each EH contains a *Next Header* field that specifies the type of the subsequent header, allowing zero or more EHs to be chained in a predefined order. This design enables the extension of packet semantics without modifying the base IPv6 protocol.

RFC 8200 [16] and the Internet Assigned Numbers Authority (IANA) [4] define a wide range of EHs. In this study, we focus on three header families that are central to our measurements and attack construction: (i) Destination Options Headers, which support in-network measurements and experimental extensions [7, 18]; (ii) Routing Headers, which (analogous to IPv4 loose source routing) can steer packets through specified intermediate nodes before reaching their final destination [12, 13, 19, 20]; and (iii) Fragment Headers, which enable source fragmentation when packets exceed the path Maximum Transmission Unit (MTU). Additional standardized EHs and the configurations used in our measurements are described in Appendix A.

Processing Complexity. The flexibility of EHs introduces significant processing overhead and complexity. RFC 8200 specifies that, with the exception of the Hop-by-Hop Options header (HBH), EHs should only be processed by the destination host. However, in a tunneled environment, the destination of the outer packet is the tunnel exit point (i.e., the

decapsulator). Crucially, middleboxes and tunnel hosts often exhibit inconsistent behavior when handling deeply nested EHs, potentially leading to slow-path processing or exception handling [33] that we exploit in TED.

2.3 Pulsing Denial-of-Service Attacks

Pulsing Denial-of-Service attacks represent a distinct evolution from continuous flooding, characterized by periodic, short-lived traffic bursts intended to evade rate-based detection. Section 7 provides a comprehensive review of prior PDoS attacks.

Temporal Lensing. The theoretical basis of high-intensity PDoS attacks is the *temporal lensing*, a theory formally proposed by Rasti et al. [57]. Similar to how an optical lens converges light rays, this technique focuses dispersed low-bandwidth data packet flows onto the victim, forming a high-density pulse. Fundamentally, the arrival time of a packet depends on its transmission timestamp plus the total network delay. Attackers can actively manipulate this delay—either by exploiting natural path delays as proposed in [57], or by artificially combining delays, as in TED, to calculate precise transmission schedules. This synchronization mechanism ensures that packets sent within a time interval arrive simultaneously at the target link within a microsecond-level time window.

Pulse Amplitude. The effectiveness of a PDoS attack is measured by the *pulse amplitude*, defined as the ratio of the peak packet reception rate within the pulse window to the attacker’s average sending rate during the accumulation window. Existing attacks like DNSBomb [45] achieve this by exploiting protocol timers (e.g., DNS timeouts). In contrast, TED achieves a PDoS attack by scheduling the forwarding paths of the packet containing EHs.

3 Extension Header Processing Analysis

The key aspect of TED lies in its ability to intentionally manipulate packet forwarding paths and cumulative delays using IPv6 Extension Headers. We will analyze the processing behavior of tunnel hosts under various EH configurations.

Tunnel Blind Spots. Our measurements reveal that most tunnel hosts adopt a lightweight *decapsulate-and-forward* model. When packets contain no extension headers, tunnel hosts typically strip all encapsulation layers and directly forward the innermost IPv6 packet toward its destination, regardless of the number of tunnel layers involved. In contrast, when extension headers are present, tunnel hosts tend to process encapsulated packets in a layer-by-layer manner, peeling off outer IPv6 headers sequentially before forwarding.

Crucially, we observe that tunnel hosts rarely perform DPI on inner packets beyond the minimum required for forwarding. This behavior creates a processing *blind spot*, where complex or deeply nested EH structures propagate through

multiple tunnel layers with limited validation, without triggering filtering or normalization mechanisms.

Exploitable Header Types. We focused on EHs that affect how tunnel hosts parse data packets. Our experiments identified the following EH types as practically exploitable in TED scenarios:

- **Fragment Header (FH):** Our measurements reveal that most tunnel hosts lack strict filtering for fragmented traffic, allowing FH packets to evade Access Control Lists (ACL) that typically inspect only initial headers.
- **Routing Header (RH):** Despite known security risks, many hosts forward packets containing RH (e.g., Segment Routing or Type 0) without verification.
- **Destination Options Header (DOH):** Tunnel hosts frequently forward DOH packets indiscriminately during decapsulation, resulting in high survivability.

Conversely, protocol-specific headers, Authentication Header (AH), Encapsulating Security Payload (ESP), and Mobility Header (MH), undergo strict processing, while Hop-by-Hop Options (HBH) are actively dropped by intermediate routers, rendering them unsuitable for TED attacks [25, 33].

Our tests indicate that approximately 70.9% of the discovered tunnel hosts forward packets containing these complex, nested EHs without filtering, allowing us to chain multiple tunnels and headers to customize the delay $D(p)$ (used in Section 4.5.1) for each packet.

4 Construction of TED Attack

In this section, we detail the construction and implementation of the TED attack, adhering to the workflow established in Section 4.2. By leveraging tunnel encapsulation, complex Extension Header processing, and path latency heterogeneity, we demonstrate that TED generates high-intensity pulsing DoS traffic with a pulse amplitude exceeding $180\times$. Furthermore, we propose a dynamic greedy pruning algorithm for efficient path selection and packet scheduling. This method reduces computational complexity while maximizing pulse fidelity by systematically discarding suboptimal path combinations during the construction phase.

4.1 Threat Model

This section defines the threat model for TED, including assumptions about the attacker, the network, and the target environment. The model reflects real-world Internet conditions and emphasizes that TED exploits protocol-level and architectural inconsistencies rather than requiring large-scale distributed resources.

Attacker. We assume an attacker controls one or more Internet-connected hosts with root privileges, enabling the

generation of raw packets with arbitrarily nested tunnel hosts and IPv6 EHs. We further assume NTP-level clock synchronization across participating hosts (millisecond precision).

Network Assumptions. Based on our measurements, a significant fraction of tunnel hosts lack strict SAV, allowing the attacker to inject packets with spoofed outer source addresses. To solve the send-time schedule problem, the attacker must estimate the one-way delay (OWD) from a tunnel host to the victim. Because direct OWD measurement is infeasible without receiver cooperation, we model the true OWD as

$$OWD = \frac{RTT}{2} + \delta, \quad (1)$$

where δ captures path asymmetry and measurement error. TED uses $RTT/2$ as the scheduling estimate, while Section 5.4 evaluates how representative δ ranges affect pulse convergence.

We justify this assumption based on three factors:

1. **Flow Label Consistency:** According to RFC 6437 and 6438 [2, 3], routers may use IPv6 Flow Labels for load-balanced forwarding, especially when encapsulation obscures transport headers. By fixing Flow Labels, packets within a flow are more likely to follow consistent paths.
2. **Experimental Verification:** Using a custom `traceroute6` tool that sends packets carrying EHs, we compare the forward path with the return path and observe high path consistency when the packet format and Flow Label are fixed.
3. **Error Tolerance:** TED targets a predefined receive time window for traffic convergence and tolerates moderate path asymmetry without disrupting pulse formation.

Victim. The victim is any publicly reachable IPv6 host or gateway with finite access bandwidth. No special tunneling configuration is required because the victim receives only the decapsulated inner payload.

Goal. The primary objective is to launch a PDos attack that maximizes the pulse amplitude. By concentrating traffic into a precise target time window, the attacker aims to induce packet loss, bypass threshold-based traffic monitoring, and cause service disruption at the victim’s link.

4.2 Attack Overview

The TED attack operates through a systematic pipeline comprising six distinct phases, as illustrated in Figure 1.

Phase 1: Tunnel Discovery Scanning. The attacker initiates the process by conducting Internet-wide scanning to identify a dataset of vulnerable tunnel hosts (Section 4.3). These hosts act as unwitting relays capable of decapsulating and forwarding ICMPv6 probe packets.

Phase 2: Path Delay Measurement. Using ICMPv6 probe packets with specific extension headers, the attacker simulates

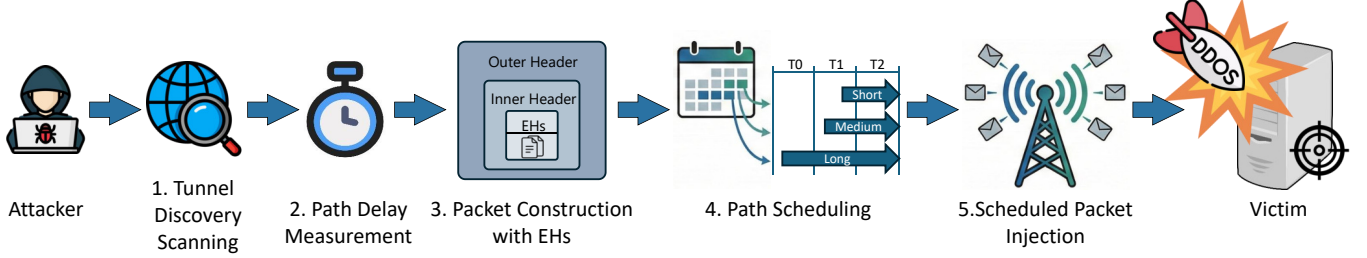


Figure 1: TED attack workflow

packet forwarding under attack conditions, measuring the RTT between the prober and the tunnel hosts, between the tunnel hosts themselves, and from each tunnel host to the victim (Section 4.4).

Phase 3: Packet Construction with EHs. Based on measured delays, the attacker constructs a set of forwarding paths $\mathcal{P} = \{p_1, p_2, \dots, p_n\}$. For each path p_i , we adjust the number of tunnel nesting layers and the combination of EH to fine-tune its total delay $D(p_i)$ (Section 4.5.1). This creates a packet transmission delay table covering a wide range of times.

Phase 4: Path Scheduling. The attacker performs a traffic schedule operation, calculating the specific release time for each packet to ensure simultaneous arrival at the target time T_{target} (Section 4.5). The transmission schedule is inverted by prioritizing packets on long-delay paths and deferring the transmission of packets on short-delay paths.

$$T_{send}(i) = T_{target} - D(p_i) \quad (2)$$

Phase 5: Scheduled Packet Injection. The attacker sends packets to the victim following the computed schedule (Section 4.5). During this phase, the transmission rate remains low and evenly distributed over a wide time window.

Despite packets having different delays, precise packet synchronization is achieved by combining IPv6 EHs with timed injection schedules. The dispersed packets converge at the victim within the microsecond-scale window, producing a high-intensity PDoS attack.

4.3 Vulnerable Tunnel Host Measurement

The core prerequisite for TED is the availability of open tunnel hosts that blindly decapsulate and forward traffic. To assess the scale of this attack surface, we conducted an active scan of the IPv4 address space and the IPv6 Hitlist [22, 59, 60, 65].

Scanning Methodology. Drawing upon prior methods for tunnel discovery [5, 29], we employ two distinct probing strategies tailored to the specific encapsulation behaviors of different tunnel protocols.

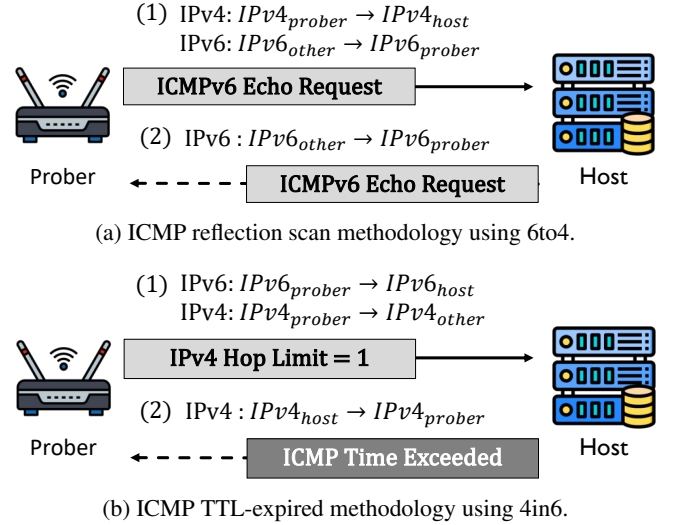


Figure 2: Illustration of the two primary scanning methods.

4.3.1 Method 1: ICMP Reflection Scan (IPIP, GRE, IP6IP6, 6to4)

For standard IPv4-based tunnels (IPIP, GRE) and IPv6-native tunnels (IP6IP6, GRE6), we utilize an ICMP Reflection Scan. Figure 2a illustrates the 6to4 instantiation of this method; the same reflection principle is universally applicable to other tunnel protocols, contingent upon the correct configuration of their respective outer and inner headers.

Specifically, the outer header targets the tunnel host's IP, while the inner header sets the *Destination Address* to the scanner's IP and the *Source Address* to the tunnel host's IP. For 6to4 tunnels, if the specific inner IPv6 address is unknown, we utilize an arbitrary remote IPv6 address ($IPv6_{other}$) as the inner destination. To prevent inbound SAV at the measurement vantage point, $IPv6_{other}$ is selected from outside the scanner's local network prefix. Upon receiving the probe, a vulnerable host decapsulates the outer header and forwards the inner packet. Receipt of this forwarded packet confirms the host's decapsulation capability.

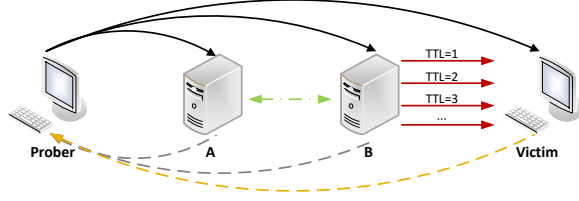


Figure 3: Delay measurement

4.3.2 Method 2: ICMP TTL Expiration Scan (4in6)

To accommodate 4in6 hosts that suppress or ignore ICMP Echo requests [5], we employ this method to verify decapsulation capabilities. This method relies on the hop-limit processing logic of the tunnel interface, as depicted in Figure 2b.

In this approach, the scanner sends an encapsulated packet containing an inner IPv4 header with a Time-To-Live (TTL) value of 1. The inner destination is set to an arbitrary address distinct from the host to ensure forwarding is attempted. Upon decapsulation, a vulnerable host attempts to route the inner payload but is forced to drop the packet when the TTL decrements to zero. This expiration triggers the generation of an ICMP *Time Exceeded* message directed back to the inner source (the prober). The prober validates the discovery by inspecting the ICMP payload, which contains the expired IP header and the original 64 bits of the datagram.

4.4 Path Construction and Characterization

Constructing the TED requires precise control over the latency of each packet. This involves two steps: accurate delay measurement and packet forwarding path combination.

4.4.1 Delay Measurement.

To construct a reliable delay map, we must accurately estimate the OWD from the attacker to the victim via specific tunnel hosts. Different from previous research on network delay calculation using King’s algorithm [28], the distinct processing overheads induced by IPv6 EHs at tunnel hosts render direct OWD measurement without receiver cooperation distinctively challenging. We therefore use a prober that sends EH-laden packets for more accurate OWD, as illustrated in Figure 3.

TED uses $RTT/2$ as the scheduler’s OWD estimate. The residual error, including path asymmetry, measurement noise, and short-term delay drift, is represented by δ . To ensure this approximation’s validity, we keep the IPv6 Flow Label fixed across all probes. Our validation confirms: when packet formats and Flow Labels align, the forward path detected highly overlaps with the return path from ICMPv6 responses. This maximizes path symmetry and minimizes measurement error. Leveraging the active measurement methods detailed in Section 4.3, we probe six tunnel protocol types. By analyzing

the received ICMP Echo Reply or Time Exceeded messages, we derive the latency for specific path segments as follows:

Prober-to-Host Latency: The foundation of our delay map is establishing the baseline latency between the prober and individual tunnel hosts. For any target host N_i , we transmit a single-layer encapsulated ICMPv6 probe and record the RTT_i . The OWD is derived as:

$$OWD(Prober, N_i) \approx \frac{1}{2} RTT_i, \quad (3)$$

where $D(Prober, N_i)$ represent the estimated link delay between *Prober* and N_i .

Host-to-Host Latency: To measure the latency between two specific tunnel hosts (e.g., Host A and Host B), we employ a differential measurement strategy. Leveraging the forwarding capability of nested tunnels, we construct a double-encapsulated packet that traverses the triangular path: $Prober \rightarrow A \rightarrow B \rightarrow Prober$. Let RTT_{ALL} be the total traversal time. By subtracting the known delays of the $Prober \rightarrow A$ and the $B \rightarrow Prober$ from the total latency, we effectively isolate the host-to-host latency $D(A, B)$:

$$D(A, B) \approx RTT_{ALL} - D(Prober, A) - D(B, Prober), \quad (4)$$

where $D(A, B)$ denotes the estimated one-way segment delay between tunnel hosts A and B.

Host-to-Victim Latency: We utilize incremental TTL probing to determine the latency from the final tunnel host to the victim. The attacker transmits encapsulated packets to the tunnel host; upon decapsulation, the inner packets are forwarded to the victim with incrementally increasing Hop Limits (1, 2, 3, ...). The path traversal concludes upon receipt of an ICMPv6 *Time Exceeded* message from a gateway router or the target. The OWD from the tunnel host A to the victim V is derived by subtracting the known path duration from the total probe time:

$$OWD(A, V) \approx RTT_{ALL} - D(Prober, A) - D(V, Prober). \quad (5)$$

These estimates are used only for packet scheduling and may deviate from the actual OWD at packet injection time. Any residual timing error caused by path asymmetry, measurement noise, or delay drift between probing and packet injection is represented by δ and evaluated in Section 5.4.

Furthermore, to ensure the successful transmission of constructed attack packets, we conducted a binary search MTU probe on each identified host to determine its maximum achievable MTU.

4.4.2 Data Packet Forwarding Path Combination

Although individual tunnel hosts provide a series of discrete basic delays, relying solely on single-hop paths is insufficient

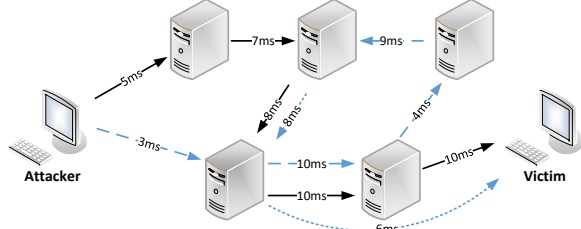


Figure 4: Schematic of TED Attack. Two packets are transmitted along distinct paths (dotted line and solid line), with edge annotations denoting the latency of each connection.

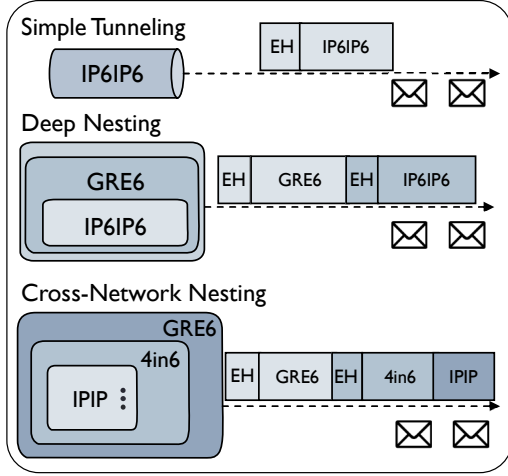


Figure 5: Schematic of Packet Construction Strategies.

to construct the continuous delay transmission schedule required for TED. As illustrated in Figure 4, we therefore adopt a **forwarding path combination strategy** with two complementary components: (i) Multi-level Tunnel Nesting, which chains multiple tunnel hosts into a composite forwarding path; and (ii) Hybrid Protocol Stacking under MTU constraints, which improves reachability by mixing heterogeneous encapsulations while discarding stacks that exceed the effective Path MTU. Collectively, these mechanisms enable the synthesis of precise transmission schedules with highly granular delay profiles.

Multi-level Tunnel Nesting: As illustrated in Figure 5, we leverage simple tunneling to implement **Deep Nesting** strategies. This approach utilizes recursive encapsulation, such as embedding GRE6 headers within outer IP6IP6 layers. To conceal the attack source, we generate a spoofed inner source address derived by cyclically adding 1 to the final octet of the outer destination address. By encapsulating a packet destined for tunnel host *B* within a packet destined for tunnel host *A*, we can construct a composite path $Prober \rightarrow A \rightarrow B \rightarrow Victim$.

Hybrid Protocol Stacking and MTU Limitations: To maximize reachability across heterogeneous networks, we employ **Cross-Network Nesting** that combines multiple tunneling

protocols (e.g., GRE6-in-IP6IP6 or 4in6-in-IP6IP6). However, the accumulation of encapsulation layers imposes strict limitations on packet sizing. Each additional encapsulation layer adds header overhead (40 bytes for IPv6 and 20 bytes for IPv4). Therefore, we only construct packets that fit within the effective Path MTU. During the construction phase, paths that violate this limitation are discarded. By systematically enumerating valid combinations of nesting and EH chains, we successfully extend the delay coverage from a discrete range of $[30ms, 500ms]$ to a continuous range of several seconds.

4.5 Dynamic Greedy Pruning Algorithm

In the preparation phase of the TED attack, a primary challenge is the combinatorial explosion of potential forwarding paths. Given approximately $N \approx 10^5$ vulnerable tunnel hosts, the permutation space for a path of depth k scales as N^k , rendering an exhaustive search for optimal delay compositions computationally intractable. A secondary challenge is “self-congestion”, where excessive traffic at intermediate tunnel hosts during the accumulation phase triggers rate limiting or buffer overflows, resulting in packet loss.

To address these constraints, we propose a **Dynamic Greedy Pruning Algorithm** for efficient path selection and schedule construction. This algorithm abstracts path latency into hop-wise components and victim-bound segments, utilizing a sorted Prior List (PL) to traverse the network topology. By dynamically pruning suboptimal or constraint-violating branches, the algorithm efficiently constructs a precise packet transmission schedule, maximizing pulse peak intensity and transmission stealth, while significantly reducing the computational complexity associated with path selection.

4.5.1 Delay Abstraction

To simplify the path-finding process, we decompose the total path latency into two atomic operations:

- τ_{next} : The latency from the current host (or source) to the next intermediate tunnel host.
- τ_{victim} : The latency calculated as the sum of the delay from the current host to the last hop and the delay from the last hop to the victim.

For a path p composed of hosts $\{N_1, N_2, \dots, N_k\}$, the total delay is approximated by:

$$D(p) = \sum_{i=1}^{k-1} \tau_{next}(N_i, N_{i+1}) + \tau_{victim}(N_k, Victim) \quad (6)$$

where N_i denotes a tunnel host, which may also correspond to the attacker host.

Before expansion, we first pick the first host that serves as the penultimate hop toward the victim. Conditioned on this fixed host, each time a new intermediate host is selected from

PL, the corresponding τ_{victim} term will be changed while the cumulative prefix delay increases; by repeatedly prepending the previous hop host of the initially chosen host, we progressively elongate the path until the target accumulation window is covered.

4.5.2 Prior List Construction

To facilitate greedy selection, we pre-compute a PL. For every host u in our dataset, we construct a sorted list of potential next-hops v , ordered by the latency $\tau_{\text{next}}(u, v)$ in ascending order:

$$PL[u] = [v_1, \dots, v_m] \quad \text{s.t.} \quad \tau_{\text{next}}(u, v_i) \leq \tau_{\text{next}}(u, v_{i+1}) \quad (7)$$

This pre-sorting enables the algorithm to iterate through the most promising candidates first and terminate search branches early once delay constraints are exceeded.

4.5.3 Algorithm Design

The algorithm 1 aims to identify a set of paths capable of covering the target accumulation window T_{accum} . To circumvent the overhead of a full Depth-First Search (DFS), we employ a *dynamic greedy expansion* strategy:

Initialization: The search process commences with a single tunnel host, directly utilizing the latency τ_{victim} retrieved from the PL.

Expansion: We iteratively extend the forwarding path to build the requisite delay. Starting from the prober (attacker), we identify the first hop and calculate the cumulative latency through this host to the initialized tunnel host and finally to the victim. Subsequently, we select a new host from the PL of the current host and insert it in front of the initialized tunnel host, recalculating the total delay. This iterative expansion continues until the cumulative delay exceeds the designated transmission time window.

Pruning: To optimize performance, we apply two pruning heuristics:

- *Upper Bound Pruning:* Due to the sorted property of the PL, if adding the smallest τ_{next} from the list exceeds the accumulation windows T_{accum} , the entire branch is pruned immediately.
- *Congestion Pruning:* To prevent self-congestion, hosts flagged as saturated by internal load counters are skipped.

Scheduling Strategy: To maximize the pulse amplitude, we employ a *Reverse-Order Scheduling* strategy. The constructed paths are sorted in descending order of total latency. Transmission initiates with the longest-delay paths and concludes with the shortest-delay paths. This scheduling ensures that the attacker's egress rate remains relatively low and constant, while the traffic convergence at the victim approximates a high-intensity pulse.

Algorithm 1 Dynamic Greedy Pruning Path Selection

Require: Prior list PL , source node s , accumulation window T_{accum} , maximum depth K , rate limit L

Ensure: Set of valid paths $\mathcal{P}_{\text{valid}}$, schedule map S

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1:  $\mathcal{P}_{\text{valid}} \leftarrow \emptyset$ 
2:  $S \leftarrow$  map initialized to 0  $\triangleright$ packet count per time slot
3:  $Q \leftarrow \{(s, 0, \emptyset)\}$   $\triangleright$ tuple: node, delay, path
4: while  $Q$  is not empty do
5:   Pop  $(u, d_{\text{curr}}, \text{path})$  from  $Q$ 
6:    $D_{\text{path}} \leftarrow d_{\text{curr}} + \tau_{\text{victim}}(u)$   $\triangleright$ victim latency from  $u$ 
7:   if  $D_{\text{path}} \leq T_{\text{accum}}$  then
8:     if  $S[D_{\text{path}}] < L$  then
9:       Add  $(\text{path} + [u])$  to  $\mathcal{P}_{\text{valid}}$ 
10:       $S[D_{\text{path}}] \leftarrow S[D_{\text{path}}] + 1$ 
11:    end if
12:  end if
13:  if  $\text{length}(\text{path}) < K$  then
14:    for each neighbor  $v$  in  $PL[u]$  do
15:      if ISSATURATED( $v$ ) then
16:        continue  $\triangleright$ skip congested nodes
17:      end if
18:       $d_{\text{new}} \leftarrow d_{\text{curr}} + \tau_{\text{next}}(u, v)$ 
19:      if  $d_{\text{new}} > T_{\text{accum}}$  then
20:        break  $\triangleright$ PL sorted; remaining  $v$  will also fail
21:      end if
22:      Push  $(v, d_{\text{new}}, \text{path} + [u])$  to  $Q$ 
23:    end for
24:  end if
25: end while
26: return  $\mathcal{P}_{\text{valid}}$ 

```

This algorithm drastically reduces the search space. By iterating through the sorted PL , we ensure that only viable path extensions are explored, enabling the construction of the attack topology in linear time relative to the number of valid edges, rather than exponential time.

5 Evaluation of TED Attacks in Real-World

In this section, we provide a comprehensive assessment of the attack surface of TED and its impact on practical applications. We focus on the scale of vulnerable hosts, interactions between different protocols, transmission characteristics, and specific case studies, including satellite networks. Our goal is to answer the following research questions:

RQ1 (Feasibility): Can TED achieve multi-hop forwarding and concentrate traffic into a single high-amplitude pulse using only tunnel hosts and extension headers?

RQ2 (Amplitude): What is the achievable pulse amplitude?

RQ3 (Efficiency): How effective is the pruning algorithm in reducing packet loss and ensuring pulse fidelity?

5.1 Experimental Setup

To empirically evaluate the TED attack while adhering to strict ethical standards, we refrained from targeting operational service hosts within the live network. Instead, we deployed six controlled, vulnerable dual-stack servers to serve as victims. Geographically distributed across China (2), Japan (1), Germany (1), and the United States (2), these targets enable the collection of latency data that faithfully reflects the complexity and heterogeneity of real-world network environments. We used these geographically distributed targets to evaluate whether our OWD symmetry and stability assumptions generalize across different global regions.

To ensure the validity and stability of our experiments, we enforced rigorous transmission constraints:

MTU Compliance: We conducted Path MTU discovery for every attack vector. Packet sizes were strictly capped based on these measurements to prevent intermediate fragmentation or packet loss.

Congestion Avoidance: To mitigate the risk of self-congestion at tunnel hosts and avoid triggering intrusion detection systems, we imposed strict usage constraints. Specifically, the attacker is limited to sending packets per host at a maximum of 10 packets per millisecond.

To quantify the performance gain over the baseline, we define the Pulse Amplitude (PA) as:

$$PA = \frac{P_r}{P_s}, \quad (8)$$

where P_r represents peak packet reception rate by the victim within the critical 20ms pulse window, and P_s is the attacker’s average send rate over the entire sending period. The 20ms pulse window is chosen to accommodate typical network delay jitter, which is experimentally observed to be below this threshold.

Experiments were conducted with varying numbers of hosts to assess scalability. Each scenario was repeated ten times to compute the mean PA and standard deviation, ensuring statistical reliability.

5.2 Dataset and Vulnerable Host Statistics

Our global measurement revealed a massive infrastructure of vulnerable hosts functioning as unintentional relays. Data collection spanned from June 2025 to January 2026, during which we conducted 44 scanning iterations at randomized intervals to ensure temporal diversity. In each scan, the probing scope covered approximately 3.7 billion IPv4 addresses and 20 million IPv6 Hitlist addresses [22, 60, 65].

The cumulative results, detailed in Table 1, indicate a widespread vulnerability across the Internet. We successfully identified a total of **1,909,440** vulnerable tunnel hosts capable of supporting the multi-hop forwarding mechanism required for the TED attack. This dataset comprises 1,618,576 IPv4 addresses and 290,864 IPv6 addresses.

Table 1: Statistics of Vulnerable tunnel hosts Discovered

Tunnel Protocol	Count
6to4	552,762
4in6	80,052
GRE	539,237
GRE6	4,681
IPIP	526,577
IP6IP6	206,131

These hosts are widely distributed geographically and belong to different organizations (Appendix B), providing a diverse set of paths with varying natural latencies, which is beneficial for constructing our TED attack.

5.3 Attack Variants: Baseline vs. TED Pulse

In this subsection, we utilize the IP6IP6 tunnel protocol as the primary tool for evaluation. Specifically, we encapsulate ICMPv6 Echo Request messages within IP6IP6 tunnels to establish a controlled experimental baseline. This configuration enables a rigorous comparative analysis between our proposed TED attack and standard PDoS, evaluating their attack effects without relying on external application-layer intermediaries or protocol-specific pulse-generation mechanisms.

This configuration enables a rigorous comparative analysis between TED and standard PDoS, evaluating their attack effects without relying on external application-layer intermediaries or protocol-specific pulse-generation mechanisms.

5.3.1 Baseline: Single-Hop Pulsing Attack

First, we established a baseline using a traditional single-hop pulsing attack [5]. We selected a set of IP6IP6 tunnel hosts exhibiting naturally occurring RTTs uniformly distributed between 30 ms and 1400 ms. The attacker transmitted 11,000 packets ordered by decreasing latency, utilizing each host exactly once, with the transmission rate capped at 10,000 packets per second (pps).

As shown in Figure 6, the arrival of packets received by the victim was relatively concentrated. More than 10,000 packets arrive in 0.15 seconds, with the majority arriving in 0.1 seconds. This formed a distinct pulse, with a pulse amplitude of $12\times$. This confirms that simple tunneling techniques provide limited temporal concentration.

5.3.2 TED Pulse: Multi-Hop with EHs

To evaluate the full-scale efficacy of the TED attack, we orchestrated a scenario utilizing multi-hop paths constructed via nested Extension Headers. Leveraging the discovered network topology, we selected a diverse set of forwarding paths to generate a broad spectrum of latency values. The experiment

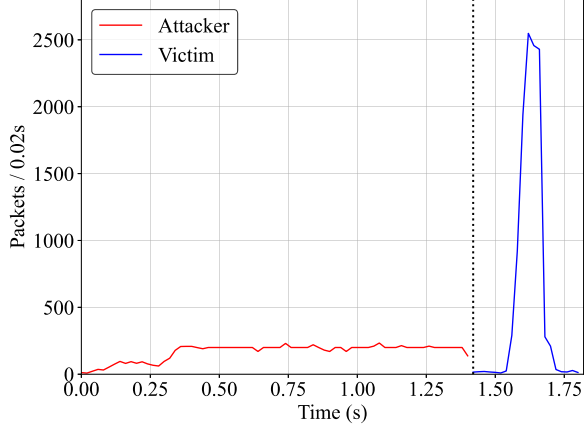


Figure 6: Baseline Scenario: Traffic convergence using naturally distributed single-hop latencies.

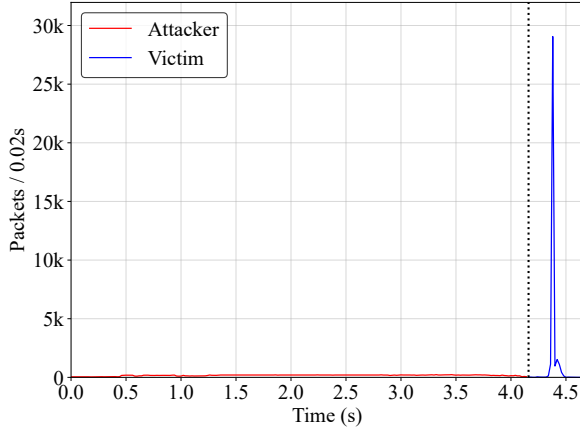


Figure 7: TED Scenario: High-intensity pulse generation via multi-hop accumulation and EH-induced delays.

employed an accumulation window of $T_{accum} = 4.2$ s, during which the attacker injected 35,029 packets. These packets were distributed uniformly across the window, corresponding to an average transmission rate of approximately 8,340 pps, with the scheduling algorithm calibrated to synchronize their convergence at the victim at time T_{target} .

The efficacy of the TED attack is illustrated in Figure 7. Consistent with the baseline scenario, the packet stream distributed over the 4.2 s accumulation window achieved successful temporal convergence at the victim. Specifically, we observed a high-density concentration of over 34,000 packets arriving within a narrow 0.25 s interval, manifesting as a sharp, distinct pulse at T_{target} . This precise alignment corroborates the ability of nested Extension Headers to enforce substantial, controllable forwarding delays across multi-hop paths. Consequently, the peak arrival rate exceeded the average injection rate by a substantial margin; specifically, with the reception window ϵ set to 20 ms, the effective pulse amplitude surpassed $180\times$. These findings confirm that TED can

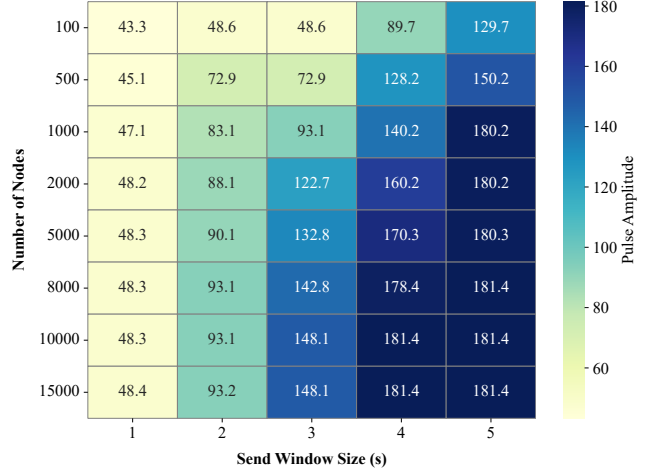


Figure 8: Heatmap of Pulse Amplitude Values varying with Number of Tunnel Host and Send Window Size.

effectively transmute low-rate, stealthy transmission streams into formidable, link-saturating traffic pulses.

Furthermore, to systematically quantify the impact of the available tunnel host population and the duration of the transmission window on attack potency, we conducted a grid search experiment while maintaining other parameters constant. The resulting pulse amplitude values are visualized as a heatmap in Figure 8.

As observed in Figure 8, the results reveal two critical insights regarding the scaling properties of TED. First, the *PA* exhibits a strong positive correlation with the *Send Window Size*. For a fixed host number, extending the window from 1-5 seconds effectively quadruples the pulse amplitude, rising from $48.3\times$ to a peak of $181.4\times$. This confirms that wider accumulation windows allow for denser packet convergence. Second, regarding the *Number of Hosts*, the *PA* initially increases rapidly as the resource pool expands from 100 to 2,000 hosts. However, the performance gains exhibit diminishing returns; beyond approximately 5,000 hosts, the *PA* saturates.

This phenomenon indicates that once the host diversity is sufficient to construct the required continuous delay spectrum, adding more hosts increases path diversity but does not significantly enhance the peak pulse magnitude.

5.4 OWD Error Sensitivity and Stability

We evaluate the impact of OWD estimation error using the model $OWD = RTT/2 + \delta$, where δ represents the total timing error observed at packet injection time. This error includes both path asymmetry in the $RTT/2$ estimate and delay drift between the probing phase and the attack phase. For a 20 ms convergence window centered at the target arrival time, a packet remains within the intended pulse window only when

Table 2: Survivability of EHs in IPv6 Tunnels

Tunnel	FH	RH0	RH2	RH4	DOH
GRE6	65.0%	64.6%	64.8%	65.4%	64.9%
IP6IP6	68.1%	71.4%	72.6%	72.2%	71.1%

$|\delta| \leq 10$ ms.

Across the six geographically distributed controlled targets, the combined OWD-error distribution shows that 82.5% of observed OWD estimation errors fall within 10 ms. We summarize this key result here and report the full CDF in Appendix C (Figure 10). This result supports our use of a 20 ms convergence window in the overall evaluation.

This total-error view also captures OWD stability over time. If the OWD of a path changes after probing, the change appears as a larger δ at packet injection time. Such drift does not invalidate TED’s path-construction mechanism, but it reduces pulse fidelity: packets whose total timing error exceeds the pulse-window tolerance arrive too early or too late and therefore do not contribute to the intended pulse.

In TED, most prober-to-host and host-to-host delay measurements are victim-independent and can be cached, while the final host-to-victim segment is refreshed before each experiment. Paths with unstable recent samples, packet loss, or latency variation exceeding the pulse window tolerance are excluded from the final schedule. Residual drift, therefore, reduces pulse fidelity and the achievable pulse amplitude by moving packets outside the target window.

5.5 Characterizing the Attack Surface

5.5.1 Protocol Interaction and EHs Survivability Analysis

To rigorously delineate the boundaries of the attack surface, we conducted comprehensive combinatorial testing. Guided by recent empirical studies on the survivability of IPv6 EHs in the wild [33], and the rate at which packets carrying EHs are correctly forwarded/decapsulated and receive normal responses from tunnel hosts, we prioritized header types with proven survivability. Specifically, we systematically examined the interaction between two native IPv6 tunneling protocols and five distinct EH variants: FH, DOH, and three types of RH. This experiment focused on quantifying the packet survival rates and processing behaviors of these specific protocol-EH combinations within the target infrastructure.

Table 2 reveals a substantial attack surface, with EH survivability averaging exceeding 70.9% across IPv6 tunnel hosts. IP6IP6 demonstrates superior permeability compared to GRE6, likely due to its architectural integration as native traffic. Structurally, GRE6 exhibits uniform survivability rates, whereas IP6IP6 displays selective filtering of FH (68.1%), reflecting the prevalence of fragmentation security policies in

native IPv6 processing.

In addition to native IPv6 protocols, we extended our combinatorial testing to include dual-stack transition mechanisms, specifically 4in6 and 6to4, in conjunction with the Extension Headers. Our results indicate that packet survivability in these dual-stack environments is significantly lower than in native IPv6 networks. This discrepancy is attributed to the maturity of the IPv4 ecosystem, which typically employs more rigorous detection mechanisms and stricter access control policies that aggressively filter anomalous packets.

Concurrently, we evaluated the efficacy of **Nested Tunneling** configurations (e.g., GRE6-in-IP6IP6, IPIP-in-4in6). While increasing encapsulation depth introduces additional processing overhead and latency jitter due to cross-domain transmission, we confirmed that nesting depths of 4-5 layers remain practically feasible. Notably, comparative analysis reveals that pure IPv6 nested structures exhibit superior performance metrics, higher packet delivery ratios and lower latency error margins—relative to hybrid nesting scenarios involving IPv4 protocols. This finding further validates the architectural advantage of exclusively leveraging the native IPv6 infrastructure for constructing stable attack paths.

5.5.2 Vulnerability Analysis and Filtering Policies

We analyzed the security policies of these hosts regarding EH processing.

Lack of Access Control: Our measurements reveal that an average of **70.9%** of the discovered IPv6 tunnel hosts have no access control strategy for packets containing Extension Headers. They blindly decapsulate and forward these packets without filtering.

Latency Discrepancy: In **81.6%** of the cases, the insertion of EHs resulted in a significant increase in processing latency compared to standard packets. This divergence underscores the inadequacy of traditional topology estimation techniques [28], which fail to account for EH-induced overhead. Consequently, this validates the necessity of our tailored methodology, utilizing EH-laden probes, to obtain the high-precision delay metrics required for the TED attack scenario.

Hidden Host Discovery: Interestingly, when performing delay measurements, we found that **0.2%** of the addresses did not respond to standard probes, but became responsive after adding EHs. We speculate that these hosts or intermediate routers may have implemented protection policies, but these policies failed to handle complex headers, ultimately revealing the hosts’ existence.

5.5.3 Transmission Characteristics

In the TED attack architecture, extending the accumulation window inherently necessitates the construction of longer forwarding paths to induce greater delays. However, empirical observations reveal a divergence between the theoretical

Table 3: Impact of Hop Count on Packet Reception Rate

Hops	Sent	Received	Reception Rate (%)
1	20,000	19,842	99.21
2	17,677	17,297	97.85
3	19,032	18,111	95.16
4	10,491	9,744	92.87
5	17,353	15,305	88.20
10	5,116	3,130	61.20
15	451	40	8.87

and actual pulse amplitude values, with this performance gap widening significantly as the path depth and time window expand. To investigate the root cause of this attenuation, we systematically evaluate the impact of packet forwarding hop counts on the transmission success rate.

To determine the impact of path length, we fixed all packet parameters, including payload size and header composition, while exclusively varying the forwarding hop count. We used stable forwarding routes validated during our initial measurement phase. Specifically, we randomly sampled 10% of the real-world paths from the dataset utilized in the TED evaluation (Section 5.3) and applied a 5 s timeout to all transmissions. We repeated the experiment for each hop depth; Table 3 summarizes the resulting reception rates. As shown, the reception rate gradually decreases as the hop increases, making packet loss more likely on longer forwarding paths.

5.6 Case Study: Threat to Satellite Networks (Starlink)

A particularly significant finding is the exposure of critical space internet infrastructure within the scanned dataset. Specifically, we identified hundreds of tunnel hosts corresponding to 3 GRE prefixes and 20 GRE6 prefixes belonging to **AS14593 (SpaceX)**, which are vulnerable to TED attacks. Reverse DNS resolution further confirmed that these addresses correspond to Starlink user terminals and ground Point-of-Presence stations [46, 50].

Despite the enforcement of strict firewall policies designed to discard unsolicited inbound traffic, we identified a critical implementation flaw wherein specific tunnel encapsulation protocols are erroneously permitted. This finding corroborates prior research [17] highlighting firewall misconfigurations within the Starlink ecosystem, thereby confirming a significantly expanded attack surface for satellite internet infrastructure. By crafting packets containing specific IPv6 EHs, we successfully circumvented these access control mechanisms, effectively weaponizing the satellite infrastructure as attack relays. This exploitation forces high-volume traffic through Starlink ground stations and user terminals, depleting scarce available uplink/downlink transmission slots, disrupt-

ing legitimate traffic transmission, and inducing substantial processing latency. Consequently, this resource exhaustion severely degrades the Quality of Service for legitimate users.

Consequently, this case study demonstrates that the threat posed by TED extends beyond terrestrial servers, endangering the stability and availability of emerging low-earth orbit satellite ecosystems.

5.7 Stealthiness Analysis

We also evaluate the operational stealth of the TED attack. During the traffic accumulation phase, the attacker maintains a low average sending rate (below 5 Mbps), causing the traffic to resemble benign IPv6 tunnel packets and thereby reducing the likelihood of triggering volumetric DDoS detectors at the attacker’s ISP edge.

TED architecture also complicates attribution. Many intermediate tunnel hosts often lack DPI for complex packets with EHs, and without parsing the inner payload. In addition, spoofing the inner source address obscures the true origin even when individual relays raise local alerts. Finally, our path-selection algorithm distributes traffic across diverse organizations and regions, preventing conspicuous concentration on any single intermediate network until the pulse is released.

5.8 Impact of Path Pruning Strategy

To quantify the effectiveness of the congestion-aware *Dynamic Greedy Pruning Path Selection* Algorithm 1, we comparatively evaluate resource consumption and attack performance with and without the pruning mechanism enabled.

Without Pruning: In the absence of path pruning, the path construction phase incurs substantially higher computational overhead, with execution time and memory usage increasing by approximately 16.8% and 31.1%, respectively. More critically, failing to eliminate potential bottleneck hosts leads to traffic collisions among multiple attack paths at intermediate tunnel hosts, resulting in self-induced congestion and buffer overflow. Consequently, the packet reception rate at the victim drops significantly, causing an attenuation of about 21.7% in the peak amplitude of the resulting attack pulses.

With Pruning: When path pruning is enabled, the algorithm proactively removes redundant branches and paths with a high risk of congestion, thereby reducing computational overhead while improving traffic distribution. Although the total number of transmitted packets is slightly reduced, the packet reception rate increases to over 94.5%. By effectively eliminating bottleneck effects at intermediate nodes, the pruning strategy produces attack pulses with steeper waveforms and higher peak intensity.

6 Discussion and Mitigation

In this section, we analyze the root causes enabling the TED attack and propose practical mitigation measures for network operators and vendors.

6.1 Root Cause Analysis

The success of TED stems from a fundamental mismatch between protocol design and device implementation in the IPv6 transition period.

The Blindness of Tunneling Logic. Tunneling protocols are designed to be stateless pipes. The forwarding logic typically focuses solely on the outer packet header to determine the next hop. The inner packet payload, including its complex EHs chain, is treated as opaque data. This architectural "blindness" allows attackers to hide inner malicious packets deep within the encapsulation, evading superficial ACLs.

Ambiguity in EH Processing Standards. RFC 8200 states that EHs should only be processed by the destination node. However, in a tunneled scenario, the definition of "destination" is ambiguous—is it the tunnel host or the final decapsulated host? Our measurements reveal that a significant proportion of intermediate tunnel hosts inadvertently process or forward packets containing EHs, creating security blind spots susceptible to exploitation.

6.2 Limitations

TED's feasibility depends on two Internet-scale timing assumptions: OWD must be close enough to $RTT/2$ for the selected pulse window, and OWD must remain stable between probing and packet injection. Although our EH-aware probing, fixed Flow Labels, multi-geolocation validation, and stability filtering reduce the impact of these assumptions, accurately evaluating OWD at Internet scale is itself challenging. Our reported attack surface should therefore be read as the fraction of measured tunnel hosts satisfying the protocol, spoofing, and timing conditions observed during our measurement period, not as a guarantee that every discovered tunnel host remains usable at all times.

TED also depends on the continued absence of strict SAV and on permissive handling of nested encapsulation and IPv6 EHs. Networks that deploy ingress/egress SAV, restrict tunnel endpoints to authenticated peers, or normalize/drop risky EH chains can substantially reduce the practical attack surface. Finally, while we conducted tests involving victims located in various regions, the exact pulse amplitude achievable against any given victim remains target-dependent because routing asymmetry, path churn, and local filtering policies vary across regions and ASes.

6.3 Mitigation Solutions

To defend against TED and similar pulsing attacks, we propose a multi-layered defense strategy.

6.3.1 Short-Term Countermeasures (Configuration)

Network operators can deploy immediate fixes using existing infrastructure capabilities:

Limiting Nested Encapsulation: Border gateways should be configured to detect and drop packets with excessive tunnel nesting levels. For legitimate traffic, a nesting depth greater than one is rare. Dropping packets with a recursion depth > 1 would neutralize the complex path construction used in TED.

Filtering Risky Extension Headers: Operators should enforce stricter policies on IPv6 EHs at the tunnel host. Specifically, blocking RH0, which is deprecated but still supported by some legacy nodes, and limiting the number of FH allowed per packet can significantly reduce the attacker's ability to manipulate path latency.

6.3.2 Long-Term Defenses (Architectural)

Regarding long-term architectural defense strategies, we propose recommendations grounded in existing research:

Enhanced SAV: A core enabler of TED is the ability to spoof the inner IPv6 source address to solicit responses or hide the attacker's identity. Widespread deployment of SAV at the tunnel ingress is crucial. If tunnel brokers verify that the inner source address belongs to the authenticated user, the attack vector is significantly constrained. Existing inter-AS SAV proposals [8, 32], together with recent deployment-oriented evaluations of deployable SAV policies [58], provide practical directions for extending source-address validation beyond a single administrative domain.

Deep Packet Inspection for Tunnels: Next-generation firewalls must extend their inspection capabilities beyond the outer header. Security middleboxes should be equipped to recursively decapsulate packets, enabling the detection of deep-layer logical anomalies such as a packet containing superfluous encapsulation layers or an inner TTL of 0.

7 Related Work

7.1 Pulsing and Low-Rate DoS Attacks

Efficiency-oriented pulsing attacks build on low-rate DoS strategies, most notably the Shrew attack [41], which exploits periodic burst traffic to target TCP retransmission timeouts. Luo et al. formalized PDoS [48] and showed that carefully selected burst patterns can significantly degrade TCP throughput while evading conventional traffic-detection mechanisms; they subsequently optimized the attack parameters to balance effectiveness and stealth [47]. To further increase pulse amplitude, Ke et al. proposed CICADAS [35], a decentralized

framework that coordinates botnet traffic to generate synchronized pulses on target paths.

Parallel to these node-centric strategies, network-centric approaches exist. The concept of *Temporal Lensing* was introduced by Rasti et al. [57], who demonstrated that natural path latency differences in the Internet could be used to focus traffic. Variants leveraging other mechanisms have emerged, including DNSBomb [45] and CDN-Convex Lens [31], which utilize intermediate devices’ buffering and aggregation logic to increase pulse amplitude.

Unlike application-layer approaches that rely on synchronizing botnet senders or relying on specific protocol dynamics, TED operates entirely at the network layer. By combining tunneling with EH-induced processing effects to construct programmable delays, TED is protocol-agnostic and reduces reliance on distributed clock synchronization, complicating mitigation via application-specific defenses.

7.2 Tunneling and IPv6 Security

Tunneling mechanisms have been extensively studied for their security implications [5,23,30,40]. Gilad et al. showed that encapsulation overhead often triggers IP fragmentation, which can increase a target’s susceptibility to denial-of-service attacks [23]. Kristoff et al. quantified the exposure by identifying over 1.5 million IPv4 hosts vulnerable to open 6in4 relay abuse [40]. Guo et al. examined 6to4 deployments and reported that more than 578k nodes remained vulnerable to address spoofing and packet injection [30]. Beitollahi et al. [5] further showed that millions of tunnel hosts behave as indiscriminate open relays, enabling IP spoofing and routing loops.

Prior work primarily emphasized natural latency effects and unauthorized forwarding. In contrast, TED exploits a deeper structural weakness: the absence of DPI in the tunneling process. By embedding complex EHs inside encapsulation, TED turns tunnel hosts into attack relays and enables multi-hop forwarding paths with artificially combined delays.

7.3 Defenses against Pulse Waves

Defenses against low-rate and pulsing attacks typically rely on detecting periodicities or statistical anomalies in traffic. Classical frequency-domain techniques (e.g., spectral analysis) exploit characteristic periodic patterns [11], while further work focuses on time–frequency analysis to detect subtler deviations [21]. More recently, fractal-based methods have been applied to traffic detection: Wu et al. [63] used Multi-fractal Detrended Fluctuation Analysis, and Tang et al. [61] proposed FR-RED, a real-time detection scheme based on fractal residuals.

To support rapid mitigation, researchers have also explored defenses implemented in programmable network planes. Alcoz et al. introduced ACC-Turbo [1], which leverages P4-

enabled switches to perform aggregate-based congestion control at line rate. In software-defined networking environments, Tang et al. [62] proposed an identification framework that jointly considers attack effectiveness and traffic characteristics, and Bocu et al. [6] incorporated federated learning to improve detection robustness in a distributed setting.

While such defenses are effective against common pulses, TED challenges these detections by spoofing tunnel source IP addresses and dynamically varying path latencies for each pulse. This high entropy in flow features and arrival times disrupts the periodicity and fractal characteristics required by current detection algorithms, rendering aggregate-based identification significantly more challenging.

8 Conclusion

In this paper, we presented **TED**, a novel pulsing Denial-of-Service attack that fundamentally shifts the paradigm of network-layer Pulsing DoS. By systematically exploiting the architectural oversight in how tunneling protocols handle nested IPv6 Extension Headers, we demonstrated that the IP tunneling infrastructure itself can be weaponized. Our research yields three key insights. First, the transitional IPv6 ecosystem is vulnerable. We identified over 1.9 million tunnel hosts (6to4, GRE, IPIP) that act as unwitting relays, processing complex header chains without adequate security validation. Second, dependency-free pulse concentration is achievable. Unlike prior attacks that rely on specific application services (e.g., DNS, NTP), TED achieves a pulse amplitude exceeding $180\times$ solely through network-layer packet manipulation. This makes the attack harder to mitigate using standard application-layer filtering. Third, the *blind* forwarding logic of current tunneling mechanisms is a major liability. The ability to hide malicious control headers deep within encapsulation layers allows attackers to bypass perimeter defenses and manipulate packet timing with high precision.

We hope that the disclosure of TED serves as a catalyst for immediate action. Mitigating this threat requires a coordinated effort: network operators must enforce stricter SAV at tunnel hosts, and equipment vendors must enhance the visibility of middleboxes into encapsulated traffic. As the Internet continues its migration to IPv6, closing these transitional loopholes is paramount to preventing a new generation of stealthy, high-impact Pulsing DoS attacks.

Acknowledgments

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Ethical Considerations

To address the ethical concerns inherent in active Internet measurements and attack simulations, we adhered to a strict code of conduct throughout this study.

This work involves Internet-scale measurements and controlled experiments related to IPv6 tunneling mechanisms, which raises important ethical considerations. Throughout this study, we strictly adhered to established ethical guidelines for network measurement research and followed best practices for responsible experimentation. We considered the main affected stakeholders to be vulnerable tunnel-host owners, infrastructure operators, controlled victim networks, and the broader Internet community that benefits from early disclosure and mitigation of this attack surface.

Measurement Scope and Minimization. Our Internet-wide measurements were limited to identifying the existence and basic responsiveness of publicly reachable IPv6 tunnel hosts (e.g., GRE6 and IP6IP6). We did not attempt to exploit vulnerabilities on individual hosts, interfere with normal traffic, or disrupt operational services. All probes were designed to be lightweight, stateless, and functionally equivalent to benign diagnostic traffic commonly observed on the Internet. Live Internet measurements were necessary because EH survivability, routing asymmetry, latency jitter, and tunnel-device forwarding behavior cannot be faithfully validated using only self-hosted testbeds.

Rate Limitation and Traffic Control. To minimize potential impact on target networks, all measurement traffic was subject to strict global and per-destination rate limits. The probing rate was conservatively bounded well below typical thresholds associated with DoS behavior and aligned with prior large-scale measurement studies. Furthermore, experiments were conducted in a distributed and randomized manner to avoid concentrating traffic on any single host or network segment. Comparative experiments without pruning were short, millisecond-scale trials with small packet volumes; any queue buildup was transient and did not impose sustained load on third-party infrastructure. For Starlink-related hosts, validation was limited to a single lightweight confirmation probe per selected address, and these hosts were excluded from all pulsing experiments.

No Uncontrolled Attack Deployment. Although this paper studies a denial-of-service attack, TED was never deployed against real victims on the public Internet. All high-intensity pulsing attacks detailed in Section 5.1 were directed exclusively at our own controlled infrastructure, ensuring that no third-party entities were targeted or harmed. For Internet-facing measurements, we focused exclusively on characterizing structural properties of tunnel hosts rather than triggering a high-magnitude pulse. We transmitted only standard encapsulation and ICMP packets, and did not send malware, exploit payloads, or application-layer attack content.

Responsible Disclosure. The vulnerabilities exposed by

TED stem from architectural and protocol-level design choices rather than isolated implementation bugs. Nevertheless, we have taken steps to responsibly disclose our findings to relevant network operators and vendors where appropriate. We aim to support mitigation efforts by providing actionable insights rather than enabling misuse. We reported Starlink-related findings to the Starlink security team and contacted coordination bodies, including CNCERT/CC, CERT/CC, UKCERT, and US-CERT. We also sent reports to major ISPs and cloud providers hosting significant numbers of vulnerable tunnel hosts, including AWS, Azure, Oracle, Google, and Alibaba Cloud.

Ethical Intent. The primary goal of this research is to improve the security of the IPv6 transition ecosystem by identifying systemic risks before they are widely exploited in the wild. We believe that transparent analysis of such structural weaknesses is essential for strengthening the long-term resilience of Internet infrastructure.

Open Science

In line with the principles of open science and reproducible security research, we are committed to making our methodology transparent and our results verifiable, while carefully balancing openness with security considerations.

We provide a detailed description of the TED attack model, experimental setup, and evaluation methodology within the paper to enable independent verification of our findings. All algorithmic components, including path construction, dynamic pruning, and scheduling logic, are specified at a level sufficient for reproduction by other researchers.

The prototype code of TED and the prototype codes of other methods mentioned in this paper (including the vulnerable tunnel host scan method via the ZMapv6 module) have been fully made available on Zenodo (<https://doi.org/10.5281/zenodo.20404927>) for relevant researchers to use, verify, and further develop.

Due to the sensitive nature of Internet-scale measurement data involving tunnel hosts and critical infrastructure, raw datasets will not be publicly released. Instead, we provide aggregated statistics, sanitized summaries, and derived artifacts sufficient to reproduce our key results without exposing specific networks or hosts to additional risk. Access to limited datasets may be granted to qualified academic researchers upon request and subject to ethical review.

Our experiments are designed to be reproducible using publicly available infrastructure and standard network measurement tools. All released artifacts are intended strictly for academic research and defensive analysis. Any commercial use, operational deployment, or malicious application is explicitly prohibited. We encourage future work to build upon our findings to develop more robust detection mechanisms and protocol-level defenses for IPv6 transition technologies.

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A Extended Description of IPv6 Extension Headers

Beyond the extension headers focused on in the main text, the IPv6 ecosystem includes several other standardized extension headers designed for security, mobility, and experimental extensions.

The **Hop-by-Hop Options** header is processed by every node along the packet’s delivery path, enabling in-network measurements and service performance monitoring [7, 18]. For network-layer security, the **Encapsulating Security Payload (ESP)** [37] and **Authentication Header (AH)** [36] form the foundation of IPsec, providing data confidentiality, integrity, and authentication.

Furthermore, the **Mobility Header (MH)** supports Mobile IPv6 by enabling signaling for node movement and reachability updates [34]. Finally, the **Host Identity Protocol (HIP)** and **Shim6 Protocol** headers address the architectural separation of identifiers and locators, facilitating multi-homing and improving resilience against path changes [51, 52].

Table 4 details the specific configurations and field settings for all IPv6 Extension Headers measured in this study.

Table 4: Description of IPv6 EHs

IPv6 EH (Alias)	Description
Destination Options Header (DOH)	The option is set to PadN, and the total EH size is 8 octets.
Fragment Header (FH)	The offset is set to zero, the M flag is set to one, the EH size is 8 octets, and the payload size is 1216 octets.
Atomic Fragment Header (AFrag)	The offset and M flag are both set to zero, and the EH size is 8 octets.
Routing Header type 0 (RH0)	The <i>Segments Left</i> field is set to one, and the corresponding address list contains the target address.
Routing Header type 2 (RH2)	
Routing Header type 4 (RH4)	
Authentication Header (AH)	The EH size is 24 octets, and all IPSec-related fields are filled with zeros.
Encapsulating Security Payload (ESP)	The EH size is adjusted according to the upper-layer payload, and all IPSec-related fields are filled with zeros.
Mobility Header type 0 (MH)	The EH size is 8 octets, and all fields are set according to RFC 6275 [34].
Host Identity Protocol Header Type 1 (HIP)	The EH size is 48 octets, and all fields are set according to RFC 7401 [51].
Shim6 Protocol Header (Shim6)	The EH size is 8 octets, and all fields are set according to RFC 5533 [52].

B Global Distribution of Vulnerable Tunnel Hosts

B.1 Global Distribution Map

The identified vulnerable IPv4 and IPv6 hosts demonstrate extensive geographic dispersion. To systematically characterize this distribution, we leveraged MaxMind’s GeoLite2 database [49] for geolocation of each IP address. Our analysis confirms a near-global attack surface, with vulnerable hosts detected in 227 out of 249 territories. As illustrated in Figure 9, this exposure transcends virtually all national borders. From the perspective of Regional Internet Registry (RIR) jurisdictions, the Asia-Pacific Network Information Centre (APNIC) accounts for the highest density of vulnerable hosts, whereas the African Network Information Centre (AFRINIC) represents the smallest proportion.

B.2 Top Countries and ASes

Tables 5 and 6 delineate the distribution of vulnerable tunnel hosts, identifying the specific nations and Autonomous Systems (ASes), highlighting where the exposed tunnel infrastructure is most heavily concentrated.

Discussion. The distribution is highly skewed, with a small number of countries and ASes accounting for a disproportionate share of exposed vulnerable tunnel hosts. This concentration indicates that targeted mitigation within these high-risk domains, such as enforcing SAV at tunnel ingress and strengthening encapsulation and EHs filtering, could substantially reduce the attack surface.

C OWD Estimation Error Distribution

The cumulative distribution of the absolute OWD estimation errors, measured across six controlled targets, indicates that

Table 5: Top countries by number of vulnerable tunnel hosts.

Country	Count
China	781,848
United States	272,609
South Korea	91,575
Singapore	90,161
Japan	86,902
Brazil	63,403
Canada	41,250
United Kingdom	34,865
India	30,685
Seychelles	29,147

for 82.5% of the measured paths, the error does not exceed 10 milliseconds (see Table 7). Since our 20-millisecond convergence window is centered on the target arrival time, this 10-millisecond bound corresponds to the permissible deviation on either side of the target time. The same distribution also shows that 94.0% of measured paths have absolute OWD estimation errors within 20 ms. These results suggest that most measured paths remain usable for pulse scheduling under the selected convergence window, while paths with larger errors are more likely to miss the intended pulse window and are therefore excluded or deprioritized during schedule construction.

To quantify how OWD error affects attack feasibility, we estimate the usable fraction under different δ thresholds using the OWD-error CDF in Figure 10. A stricter δ threshold improves pulse-window convergence but leaves fewer usable tunnel paths or hosts, which can reduce the achievable pulse amplitude.

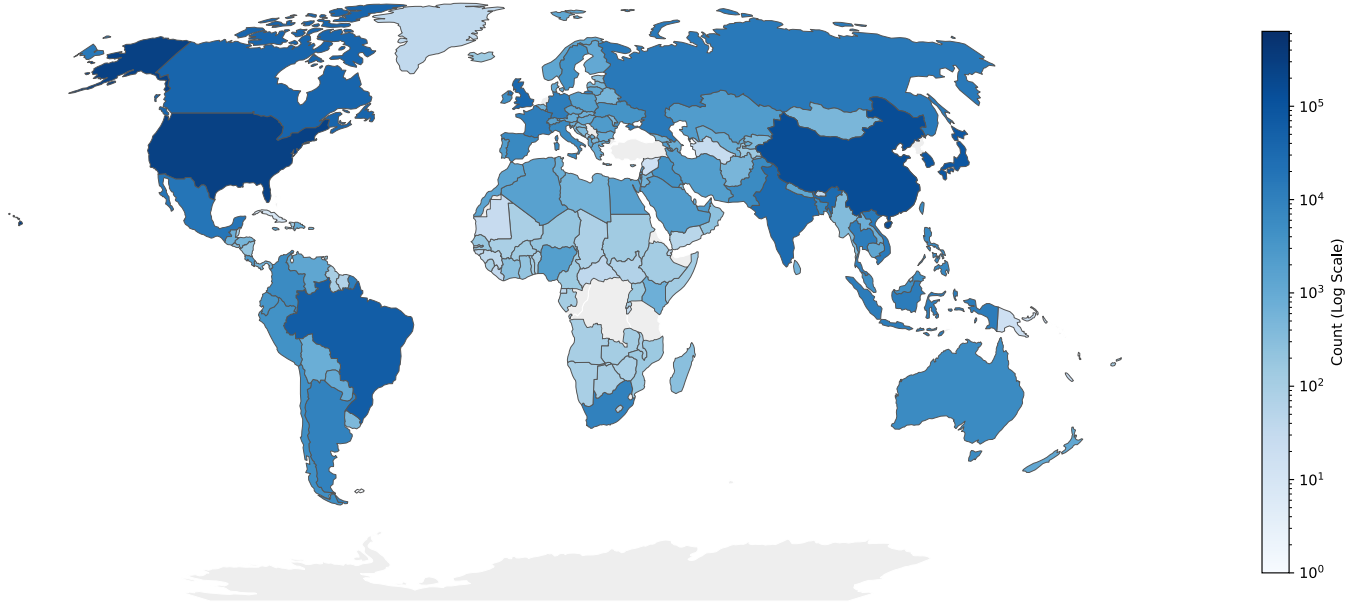


Figure 9: Global distribution of vulnerable tunnel hosts observed in our Internet-wide measurements.

Table 6: Top ASes by number of vulnerable tunnel hosts.

ASN	Count
4134	186,796
4538	106,079
139341	82,981
54600	74,799
4837	72,155
17676	71,275
9808	65,107
9381	63,351
4766	46,107
398823	36,479

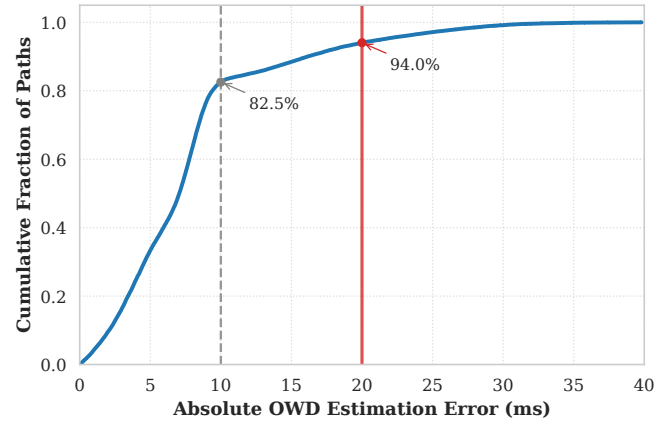


Table 7: OWD-error sensitivity of the usable tunnel-host fraction.

Timing condition	Usable fraction
$ \delta \leq 5$ ms	33.3%
$ \delta \leq 10$ ms	82.5%
$ \delta \leq 20$ ms	94.0%
$ \delta > 20$ ms	5.9% excluded or re-probed

Figure 10: CDF of absolute OWD estimation errors across six geographically distributed controlled targets. The dashed gray line marks the proportion of paths falling within the 10 ms tolerance of our convergence window, and the red line shows the proportion of paths that can successfully arrive within a full 20 ms deviation.